

For the first assignment, our group wanted to implement something rather useful for daily use, a grocery/food storage application. This application consists of three pages: the grocery list page to add items (stored using a JSON file), where the main functionality for all REST APIs is utilized to retrieve items and update item criteria, the add items page, and the analytics page that tracks budget and grocery store necessities of a family.

Understanding how the client-server interaction worked in real time, especially with GET, POST, PATCH, and DELETE requests, was also challenging at first. We had to debug several issues where data was not updating properly due to incorrect routing or not refreshing the state after modifications. Some challenges consisted of working with Chart.js, as it was initially confusing to understand how the charts were dynamically populated using data fetched from the backend. It required us to properly structure the JSON responses and ensure that the frontend correctly handled requests before rendering the charts.

However, there were many successes throughout the project. One major success was successfully implementing full RESTful functionality, allowing users to dynamically add, edit, delete, and retrieve grocery items without refreshing the page. The analytics page was also a highlight of the project, as it made the application feel more complete and made it user-friendly for navigation. Overall, we picked up a lot of insight on making the application more expandable for future use while limiting changes that may disrupt modularity.